Jacob Rosario

Software Engineer
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Professional Summary

Software engineer specializing in web, iOS, and wearable sensor applications using TypeScript, JavaScript, Python, and Swift. Mid-level professional with proven track record delivering geospatial routing and real-time sports analytics solutions, including software deployed in over 20 competitive matches. Skilled at translating user needs and complex data into responsive, practical products. Seeking to apply cross-platform development expertise to drive innovative, data-driven solutions.

Skills

- **Languages**: JavaScript, Python, Swift, HTML, CSS, C++ (prototyping)
- **Frameworks & Libraries**: React, SwiftUI, Three.js
- **Applied Engineering**: Geospatial Routing, Offline PWAs, Motion-Sensor Processing, Embedded Systems & Prototyping
- **Tools & AI**: Claude, Codex, Git, GitHub

Relevant Experience

Embedded Systems Developer (Contract) **Oct 2026 - Present**
Canadian Athlete Development *Vancouver, BC*

- Developing C++ firmware for an ESP32 wearable to process six-axis IMU data, detect jumps, and calculate jump count and average/maximum estimated height.
- Building Python simulation and C++ replay tools to evaluate detection across volleyball movements, including 4,800 simulated jumps and 9,000 non-jump scenarios, with regression coverage for sensor faults and session controls.
- Creating a React/TypeScript companion app prototype with Bluetooth Low Energy session controls and local session history, plus a standalone CSV export workflow for spreadsheet analysis.

Software Developer & Sports Analyst Volleyball Stat Tracker **Feb 2026 - May 2026**
Hustle Volleyball Club *Burnaby & Vancouver, BC*

- Built and deployed a React application used courtside on iPad across 20+ competitive matches, replacing handwritten tracking with live statistics, visual summaries, and a coaching whiteboard.
- Refactored redundant application state into a single source of truth, eliminating state drift and improving data consistency.
- Worked directly with coaches to translate match-day needs into software features and use performance data to inform lineups, practices, and in-game adjustments.

Projects

RainCouver (Rain-Cover Navigation) | React, TypeScript, LiDAR/OSM data RainCouver **Present**

- Building a PWA that routes pedestrians through Downtown Vancouver by maximizing covered walking paths, using LiDAR-derived cover data across 25,000+ street segments and 3,000+ searchable places.
- Designing a self-improving routing system where user corrections update cover data in real time without disrupting an active route, solving GPS drift and ambiguous-turn edge cases along the way.

RepPower | Swift, SwiftUI RepPower **Sep 2026**

- Developed an iOS application that uses AirPods motion-sensor data to estimate barbell squat velocity for velocity-based training.
- Engineered a Swift physics pipeline that transforms raw motion data into vertical acceleration and velocity while accounting for sensor noise and integration drift.
- Implemented zero-velocity drift correction and optimized high-frequency processing to maintain stable real-time measurements without blocking the SwiftUI interface.

Spike Timer | JavaScript, React Spike-Timer **Mar 2026**

- Built a client-side video-analysis tool that measures spike timing from user-marked takeoff, contact, and landing frames.
- Selected a lightweight geometric approach instead of pose-detection machine learning, enabling responsive browser-based processing.

Other Experience

Performance Analyst & Assistant Coach **Sep 2025 - Dec 2025**
Vancouver College *Vancouver, BC*

- Analyzed 5+ matches/week using video review and statistical data to identify opponent tendencies and performance patterns.
- Translated video data into actionable recommendations for lineups, opponent-specific strategies, and player development.

Volleyball Content Creator | @heartoverheight.vb **May 2025 - Apr 2026**

- Earned over 1 million views on multiple volleyball posts showing that shorter athletes can succeed as spikers.
- Secured sponsorship deals with Way of Wade and VKTRY Insoles

Barista **Sep 2023 - Sep 2025**
Starbucks *San Francisco, CA & Vancouver, BC*

- Processed 100+ transactions per shift at high-volume locations while maintaining accuracy and service quality under pressure.

Education

University of British Columbia **Jun 2025**
Bachelor of Science (BSc), Biology *Vancouver, BC*

- **Achievements:** Dean's List

Hobbies & Interests

Competitive volleyball, coaching & performance analysis | Sports analytics & biomechanics | AR/VR & 3D technology | Video games | Weightlifting & Olympic lifting | Content creation & video editing | Anime & manga | Playing guitar